

SBSNet: Spatial–Spectral Background–Target Separation Network for Hyperspectral Target Detection

Jianlin Xiang, Yanshan Li , Linhui Dai, Ruo Qi , Haojin Tang , Li Zhang , Kunhua Zhang , and Weixin Xie 

Abstract—Hyperspectral target detection (HTD) aims to identify target locations in a hyperspectral image (HSI) using limited prior target spectra. Existing methods often use contrastive learning to construct target and background sample sets from unlabeled HSI and compare their similarity in feature space to enhance background–target separability. However, they often fail to ensure high-purity sample sets, limiting their ability to effectively separate target and background features. Therefore, we propose a spatial–spectral background–target separation network (SBSNet). The SBSNet leverages prior target spectra to construct high-purity target and background sets from the unlabeled HSI and integrates them into a multiscale spatial–spectral feature learning framework to optimize the feature space for more discriminative target detection. Specifically, the primary contributions of this article are threefold. First, we propose a local spatial–spectral feature fusion module to extract spatial–spectral feature from the raw HSI and proposes a spatial–spectral pseudolabel purification strategy to obtain pure target and background pixel sets from unlabeled HSI. In addition, we introduce the pseudolabel map as prior information to supervise the training process. Second, we design a highly robust multiscale spatial–spectral autoencoder specifically for HTD, which is used for sample generation during the data preparation and for feature extraction during the training. Third, we propose a clustered adaptive focus training strategy, which synergistically optimizes the feature space through clustered sampling and adaptive exponential weighted loss. Finally, experimental results demonstrate that the proposed SBSNet achieves superior detection performance on five public HSI datasets in various scenarios, compared with state-of-the-art HTD methods.

Index Terms—Autoencoder (AE), contrastive learning (CL), data enhancement, hyperspectral target detection (HTD), multi-scale feature.

I. INTRODUCTION

HYPERSPECTRAL imaging technology utilizes imaging spectrometers to capture a large amount of continuous narrowband image data across a wide spectral range, from ultraviolet to far-infrared, resulting in the generation of hyperspectral images (HSIs). Compared to traditional RGB images, an HSI provides a more detailed characterization of the spectral properties of surface materials, offering a wealth of information. Due to this advantage, the HSI has been widely applied in various visual tasks, such as land cover classification [1], target detection [2], [3], and anomaly detection [4]. Among these, hyperspectral target detection (HTD) aims to identify target locations in the HSI using limited prior target knowledge, playing a significant role in fields such as aerospace [5], agriculture [6], military reconnaissance [7], marine monitoring [8], and mineral exploration [9].

Over the past three decades, HTD has been extensively studied, leading to the development of two major categories: classical methods and deep-learning-based (DL-based) methods. Traditional HTD methods primarily rely on statistical distribution or subspace models [10], [11], [12], [13], [14], [15] to model spectral differences for target detection. However, they face limitations in handling high-dimensional data and nonlinear features. In recent years, the advancement of DL has given rise to DL-based HTD methods [16], [17], [18], [19], where convolutional neural network (CNN) and attention mechanisms play a crucial role. Unlike traditional methods, DL can automatically learn complex spectral and spatial features. However, DL-based HTD methods face challenges, such as few labeled samples and class imbalance in datasets, which affect their generalization ability [20].

To address these challenges, contrastive learning (CL) has emerged as a promising paradigm for HTD. Existing CL methods can be broadly divided into two categories.

- 1) Prior-enhanced CL methods [21], [22], [23], [24], [25], [26] first extract target and background sets from unlabeled HSI using prior target spectra and then enhance the target samples to balance the target and background sets before training the network with CL. In these approaches, target and background sample sets are typically constructed by

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Our code will be available at <https://github.com/xjlDestiny/SBSNet.git>.
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selecting a fixed number of pixels with the highest or lowest similarity scores to the prior targets. As a result, only a small portion of the pixels is used to represent targets and background, and the extracted sample sets do not adequately reflect the full spectral variation in the original HSI. Consequently, these methods cannot construct high-purity target and background sample sets, which limits the discriminative capability of the learned representations [27];

- 2) self-supervised sample construction CL methods [28], [29] directly construct positive and negative sample pairs from unlabeled HSI through self-supervised learning and then train the network using CL. Due to the lack of reliable prior information constraints during sample pair construction, target and background samples are prone to being mixed, which decreases the discriminative capability of the learned features.

We consider that the key to CL lies in constructing positive and negative sample pairs that possess high purity, and in adopting effective feature optimization strategies to enhance the separability between samples and the discriminative capability of the network. Therefore, we propose a spatial-spectral background-target separation network (SBSNet) for HTD. The SBSNet employs prior target spectra in combination with a spatial-spectral pseudolabel purification strategy (SPPS) to extract high-purity background and target sets from unlabeled HSI. These sets are then integrated into CL through a multiscale spatial-spectral autoencoder (MSSAE) and a clustered adaptive focus training strategy (CAFTS), leading to more robust target detection performance. Specifically, the main contributions of this article are as follows.

- 1) We introduce a local spatial-spectral feature fusion module (LSFFM) that adaptively captures neighborhood similarity, enabling more discriminative target-background separation.
- 2) We propose the SPPS that enhances the reliability of pseudolabel maps from coarse detection via adaptive threshold, addressing the issue of low-purity sample sets in CL for the unlabeled HSI.
- 3) We propose a pseudolabel-guided supervision mechanism, which leverages pseudolabel maps as intrinsic guidance and employs progressive performance evaluation to retain optimal model states.
- 4) We design the MSSAE that seamlessly bridges data preparation and model training, providing a versatile feature representation backbone for HTD.
- 5) We propose the CAFTS, which adaptively emphasizes difficult samples during training to enhance discriminative spectral feature learning for HTD.

II. RELATED WORKS

A. Classical HTD Methods

In the initial stages of HTD, researchers predominantly relied on intuitive spectral difference principles. These methods involved assessing the spectral similarity between the target spectrum and a prior target spectrum. Common algorithms include

spectral angle mapper (SAM) [30], Euclidean distance [31], and spectral information divergence [32], which measure spectral similarity by calculating the distance, angle, and divergence between spectral vectors, respectively. However, as the complexity of HSI increased, the limitations of methods relying solely on spectral differences became apparent.

Therefore, statistical theory was incorporated, leading to a series of classical detection methods, such as the generalized likelihood ratio tests [10], adaptive coherence estimation (ACE) [11], and matched filter [12]. These algorithms approach target detection as a hypothesis-testing problem based on Gaussian distributions, significantly enhancing both detection accuracy and robustness. Farrand and Harsanyi [13] introduced the constrained energy minimization (CEM) algorithm, marking the beginning of a new phase in HTD. The CEM algorithm improves contrast between targets and backgrounds by constraining the energy of the filter output, thereby effectively separating target signals. Building on the success of CEM, Hierarchical CEM [33] and Ensemble CEM (E-CEM) [34] further enhanced detection performance, particularly in complex backgrounds.

Concurrently, orthogonal subspace projection (OSP) [14] emerged as a subspace model algorithm that constructs a subspace orthogonal to background features, thereby reducing data dimensionality and enhancing detection efficiency. Matched subspace detection [35] built upon this approach by further investigating the use of background information, thereby opening new avenues for subsequent research.

In the 21st century, the sparsity-based target detector [15] capitalized on the sparse characteristics of targets in the HSI, achieving efficient target recognition through dictionary learning and sparse coding. Building on this foundation, the combined sparse and collaborative representation [36] not only retained the principles of sparse representation but also incorporated the concept of collaborative representation, creating a comprehensive representation of targets and backgrounds to enhance detection accuracy.

Despite the notable achievements of these traditional methods, their generalization capabilities remain constrained in complex scenarios and dynamic environments. For example, these methods often rely heavily on predefined models of targets and backgrounds, resulting in suboptimal performance under novel or extreme conditions. As hyperspectral data and application scenarios grow increasingly complex, the limitations of traditional methods have become more pronounced, prompting researchers to explore more flexible and robust modern approaches, such as DL.

B. DL-Based HTD Methods

Existing DL-based HTD methods can be broadly categorized into four main paradigms: transfer learning, background learning and reconstruction, interpretable representation learning, and CL.

Transfer learning methods pioneered by CNND [37] address sample scarcity through cross-domain knowledge transfer. By pairing pixels from source HSI data to augment training samples and fine-tuning on target domains, CNND demonstrated early

success in mitigating data limitations. However, its reliance on external source data limited applicability to scenarios with domain shifts. To reduce dependence on external data, subsequent works explored synthetic sample generation.

Background learning and reconstruction methods focus on modeling background distributions using generative architectures. For instance, the background learning with target suppression constraint (BLTSC) [38] proposes an innovative semisupervised HSI target detection framework by combining adversarial training with target suppression constraints. The core idea is to accurately model the background distribution using adversarial autoencoder (AE) [39] and reduce target sample interference through multiloss joint optimization. The orthogonal subspace-guided variational autoencoder (OS-VAE) [40] introduces a variational autoencoder (VAE) integrated with OSP [14] theory. Unlike the adversarial training used in BLTSC, the OS-VAE utilizes the probabilistic latent space of the VAE to model the background distribution.

Interpretable representation learning methods bridge DL with physical models to enhance transparency. For instance, HTD-IRN [41] exemplifies this direction by combining lightweight Transformers with CEM [13] to explicitly separate target-background subspaces. By incorporating spectral separation mechanisms derived from traditional detection theory, HTD-IRN not only improves detection accuracy but also provides interpretable insights into feature representations.

CL methods are designed to learn meaningful feature representations by comparing the similarities and differences between data samples. The 3-D macro-micro residual AE [21] constructed patches of varying sizes centered on pixels in the HSI, which are then input into two separate branches for multiscale feature extraction. In addition, it incorporates a distance constraint that combines spectral and spatial information to enhance reconstruction accuracy. The distance-constraint-based stacked sparse autoencoder (DCSSAE) [22] constructed dictionaries for background using the automatic target generation process [42] and SAM [30]. By introducing distance constraints, the DCSSAE maximizes the distinguishability between target and background pixels in the feature space, thereby enhancing detection accuracy and effectively suppressing background interference. HTD-Net [23] proposed an AE architecture inspired by U-Net [43] to generate target samples and employed linear prediction for background selection. While effective in data augmentation, its use of spectral differences as input weakened discriminative features. This issue was later addressed by TSCNTD [24], which introduced a two-stream network to directly process raw spectral features and synthesized samples by mixing target priors with background pixels. Concurrently, researchers recognized the need for background suppression and band selection in complex scenes. The spectral aggregation and separation network [25] combined Target Band Random Masking with Multiple Representative Background Selection to suppress interference from heterogeneous backgrounds. Building on this, DLBSTD [26] further integrated band selection into detection via multidepth multibranch networks, jointly optimizing band redundancy and sample imbalance through Target Band Random Substitution. SCLHTD [28] introduced a method

for learning spectral differences without prior information by training adversarial convolutional AEs on odd and even bands of the HSI, thereby constructing a discriminative task for spectral differentiation. Finally, spectral similarity is utilized for detection, and edge-preserving filters are applied to refine the detection results. Chen et al. [29] proposed a spatial-spectral contrastive SSL-based HTD method with dual path networks (DPN-CSSTD) to address the scarcity of labeled data and model transferability challenges in HTD. This approach utilizes self-supervised pretraining to extract discriminative spatial-spectral features and implements transfer learning strategies to achieve cross-scenario detection capabilities.

III. PROPOSED METHOD

This section provides a detailed description of the proposed SBSNet. The algorithm consists of three phases: data preparation, SBSNet training, and SBSNet testing, as illustrated in Fig. 1.

In the data preparation phase, we propose the LSFFM to enhance the feature representation capability of raw HSI data. Building on this, we introduce the SPPS to construct background and target pixel sets from unlabeled HSI. Subsequently, the proposed MSSAE is employed to reconstruct these samples, generating a diverse and balanced training set. Following this, a clustering algorithm is applied to partition the target and background sample sets into ten clusters each. From each cluster, samples are randomly selected batch by batch to generate the training data until all samples have been used.

In the training phase, the batch-extracted data $\mathbf{X}^{\text{Batch}}$, along with prior spectral vectors $\mathbf{t}$, are input into the MSSAE for feature extraction. The parameters of the MSSAE are optimized through the adaptive exponential weighted loss (AEW-Loss) from the encoder and the reconstruction loss from the decoder.

In the testing phase, the spectral vectors of the pixels to be detected, together with the prior target spectral vectors, are input into the trained MSSAE (excluding the decoder), where feature encoding is performed. The final detection result is obtained by calculating the spectral angle similarity. In the following sections, the individual modules proposed in this article will be discussed in detail.

A. Background and Target Pixel Selection

In practical applications of the HSI, accurately obtaining ground-truth labels is often challenging, which poses significant difficulties for DL-based HTD. To enhance the feature representation capability of raw HSI data, we propose the LSFFM, which dynamically computes the SAS between adjacent pixels and the central pixel. This process preserves the spectral characteristics of the central pixel while adaptively integrating spatial information from its surrounding environment to construct a weighted spatial-spectral vector (SV). Building on this, we propose the SPPS, which employs the ACE [11] as a coarse detector to generate an initial detection map. An adaptive threshold is then applied to refine the coarse detection results, extracting a more precise pseudolabel map. This strategy enables the construction of background and target pixel sets from the unlabeled HSI.

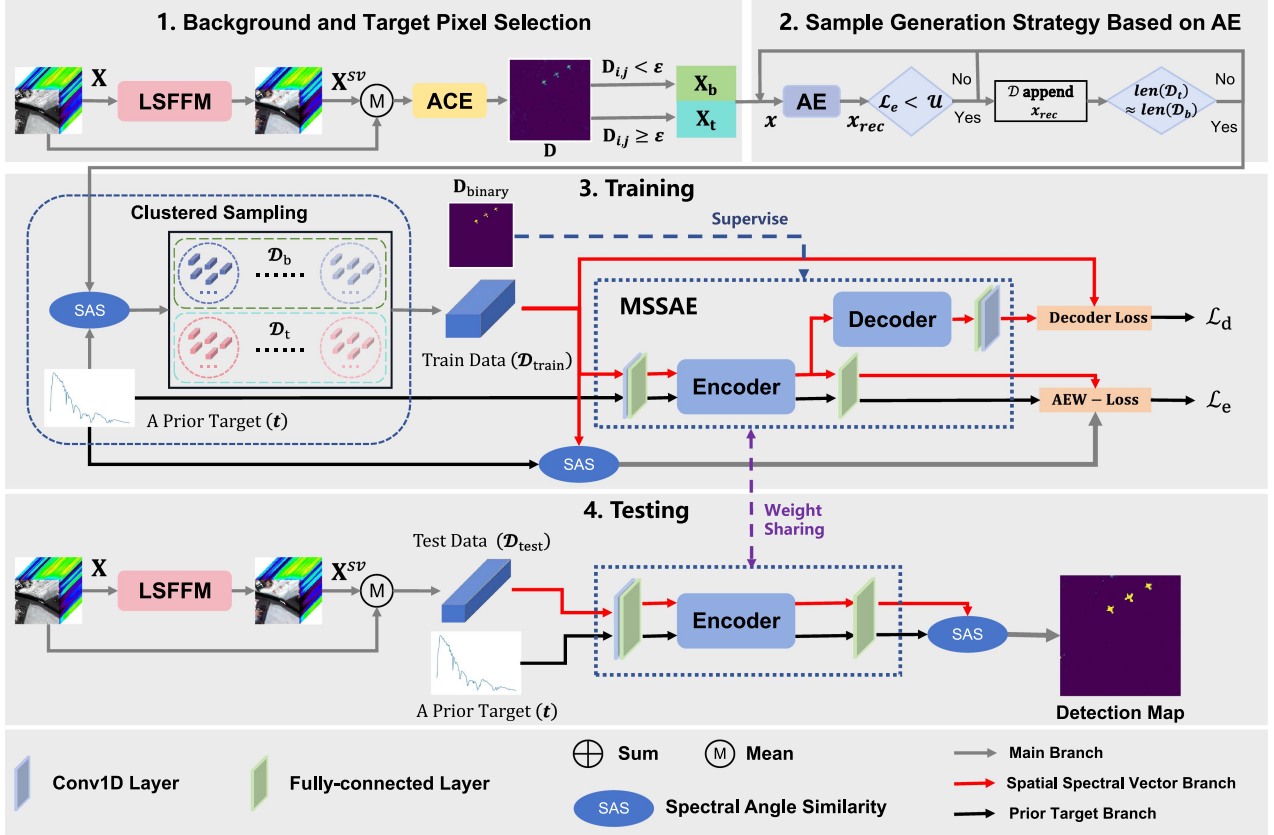


Fig. 1. Flowchart of the proposed SBSNet, which includes three phases: data preparation, SBSNet training, and SBSNet testing. D_{binary} is the pseudolabel map obtained by binarizing D using the threshold ϵ .

Let the unlabeled HSI be denoted as $\mathbf{X} \in \mathbb{R}^{H \times W \times L}$, and a priori target spectral vector as $\mathbf{t} \in \mathbb{R}^{1 \times L}$, where L represents the number of spectral bands in the HSI. First, we construct an SV for each pixel in $\mathbf{X}$, incorporating information from its neighboring pixels using the LSFFM. Given that there can be significant class differences between the central pixel and some of its neighboring pixels, a simple weighted summation of all neighboring pixels could reduce the purity of the SV of the central pixel. To mitigate this, we calculate the correlation coefficients between the neighboring pixels and the central pixel and perform a weighted summation to obtain a more refined SV. Let the spectral vector of the central pixel be denoted as $\mathbf{s}_c$, and the spectral vectors of the $S^2 - 1$ ($S > 1$ and S is an odd number) neighboring pixels in the local spatial-spectral domain as $\mathbf{s}_p$ ($0 \leq p \leq S^2 - 2$). To measure the similarity between $\mathbf{s}_p$ and $\mathbf{s}_c$, this article adopts SAS as the metric. The computation of SAS is denoted as

$$\text{SA}(\mathbf{s}_p, \mathbf{s}_c) = \cos^{-1} \left(\frac{\mathbf{s}_p^T \cdot \mathbf{s}_c}{\|\mathbf{s}_p\| \cdot \|\mathbf{s}_c\|} \right) \quad (1)$$

$$\text{SAS}(\mathbf{s}_p, \mathbf{s}_c) = 1 - \frac{\text{SA}(\mathbf{s}_p, \mathbf{s}_c)}{\pi} \quad (2)$$

where $\text{SA}(\mathbf{s}_p, \mathbf{s}_c)$ and $\text{SAS}(\mathbf{s}_p, \mathbf{s}_c)$ represent the spectral angle and spectral correlation between $\mathbf{s}_p$ and $\mathbf{s}_c$ within the local neighborhood, respectively.

To prevent the loss of spectral information caused by spatial feature descriptions, we multiply each $\text{SAS}(\mathbf{s}_p, \mathbf{s}_c)$ by $\mathbf{s}_p$ and compute the mean to obtain an SV that captures the feature information within the local neighborhood centered at $\mathbf{s}_c$, as shown in the following equation:

$$\text{SV}_p = \frac{1}{S^2 - 1} \sum_{p=0}^{S^2-2} [\mathbf{s}_p \cdot \text{SAS}(\mathbf{s}_p, \mathbf{s}_c)]. \quad (3)$$

By iteratively applying the LSFFM algorithm, the SV of all pixels in the raw HSI data $\mathbf{X}$ is computed. This results in HSI data $\mathbf{X}^{\text{sv}} \in \mathbb{R}^{H \times W \times L}$ that incorporate local spatial information.

Subsequently, the averages of $\mathbf{X}$ and $\mathbf{X}^{\text{sv}}$ are computed to obtain $\mathbf{X}^{\text{avg}} \in \mathbb{R}^{H \times W \times L}$, which is then input into the ACE detector for coarse detection, resulting in the detection map $\mathbf{D} \in \mathbb{R}^N$. The output response of ACE is denoted as

$$\mathbf{D}_{i,j} = \frac{(\hat{\mathbf{t}} \cdot \mathbf{R}^{-1} \cdot \hat{\mathbf{X}}_{i,j}^{\text{avg}})^2}{(\hat{\mathbf{t}} \cdot \mathbf{R}^{-1} \cdot (\hat{\mathbf{t}})^T) (\hat{\mathbf{X}}_{i,j}^{\text{avg}} \cdot \mathbf{R}^{-1} \cdot (\hat{\mathbf{X}}_{i,j}^{\text{avg}})^T)} \quad (4)$$

where $\boldsymbol{\mu} \in \mathbb{R}^{1 \times L}$ is the mean of all SV in $\mathbf{X}^{\text{avg}}$, $\hat{\mathbf{t}} = \mathbf{t} - \boldsymbol{\mu}$ is the de-meaned prior target vector, $\hat{\mathbf{X}}_{i,j}^{\text{avg}} = \mathbf{X}_{i,j}^{\text{avg}} - \boldsymbol{\mu}$ is the de-meaned SV of $\mathbf{X}_{i,j}^{\text{avg}}$, $\mathbf{D}_{i,j}$ is the detection response value for the (i, j) pixel position, and $\mathbf{R} \in \mathbb{R}^{L \times L}$ is the background covariance matrix.

Algorithm 1: Local Spatial–Spectral Feature Fusion Module.

Input: $\mathbf{X}$ - the input HSI, S - window size (an odd number and greater than 1)

Output: $\mathbf{X}^{\text{sv}}$ - the HSI with local spatial spectral domain correlation

- 1 Initialize the output data array $\mathbf{X}^{\text{sv}}$ with zeros of dimensions $H \times W \times L$;
- 2 Initialize window radius as $r = \lfloor S/2 \rfloor$;
- 3 Pad the data to create $\mathbf{X}_{\text{pad}}$ with dimensions $(H + 2r) \times (W + 2r) \times L$;
- 4 Assign the padded region with $\mathbf{X}_{\text{pad}}[r : H + r, r : W + r, :] = \mathbf{X}$;
- 5 **for** $i = 0$ **to** $H - 1$ **do**
- 6 **for** $j = 0$ **to** $W - 1$ **do**
- 7 Extract the local data and reshape it into vectors
 $\mathbf{D}_{\text{local}} \leftarrow \mathbf{X}_{\text{pad}}[i : i + 2r + 1, j : j + 2r + 1, :]$;
- 8 Extract the center pixel
 $\mathbf{x} \leftarrow \mathbf{X}_{\text{pad}}[i + r, j + r, :]$;
- 9 Calculate similarity weights as
 $\mathbf{W}[i] \leftarrow \text{SAS}(\mathbf{D}_{\text{local}}[i], \mathbf{x})$;
- 10 Assign weighted sum as
 $\mathbf{X}^{\text{sv}}[i, j, :] \leftarrow \text{dot}(\mathbf{D}_{\text{local}}^T, \mathbf{W}) / \sum \mathbf{W}$;
- 11 **return** $\mathbf{X}^{\text{sv}}$;

Due to the differences in the response distributions of detection maps following coarse detection across different datasets, the use of a fixed threshold may result in confusion between target and background pixels, particularly when significant differences exist between datasets. To address this, we propose an adaptive thresholding method that dynamically adjusts the threshold based on the maximum and minimum values of each dataset, thereby better accommodating variations in the background. The potential locations of target and background pixels in the HSI can be obtained using the following equation:

$$\text{POS}_t = \{(i, j) | \mathbf{D}_{i,j} \geq \varepsilon\}, \text{POS}_b = \{(i, j) | \mathbf{D}_{i,j} < \varepsilon\} \quad (5)$$

where $\varepsilon = \lceil \frac{\max(\mathbf{D}) - \min(\mathbf{D})}{k} \rceil + \min(\mathbf{D})$, and $\max(\mathbf{D})$ and $\min(\mathbf{D})$ represent the maximum and minimum values of the detection map $\mathbf{D}$, respectively, while $k \in \mathbb{Z}^+$ is a hyperparameter, which is set to 10 in all experiments presented in this article. Using the following equation, we can obtain a relatively pure initial set of background and target training samples

$$\mathbf{X}_b = \bigcup_{(i,j) \in \text{POS}_b} (\mathbf{X}_{i,j}, \mathbf{X}_{i,j}^{\text{sv}}, \mathbf{X}_{i,j}^{\text{avg}}) \quad (6)$$

$$\mathbf{X}_t = \bigcup_{(i,j) \in \text{POS}_t} (\mathbf{X}_{i,j}, \mathbf{X}_{i,j}^{\text{sv}}, \mathbf{X}_{i,j}^{\text{avg}}). \quad (7)$$

Furthermore, we introduce a pseudolabel-guided supervision mechanism. The coarse detection map $\mathbf{D}$ is binarized using the adaptive threshold ε to obtain a pseudolabel map $\mathbf{D}_{\text{binary}}$, which provides intrinsic guidance during training. After each epoch, the network output is evaluated against $\mathbf{D}_{\text{binary}}$ using the AUC

metric, and model parameters are retained only when performance improves. This progressive evaluation strategy stabilizes training under noisy pseudolabel supervision.

B. Sample Generation Strategy Based on the AE

DL has advanced computer vision significantly, benefiting from large labeled datasets. However, in the HSI, the scarcity of labeled samples and class imbalance between background and target categories pose major challenges. To address these challenges, this article innovatively designs a sample generation strategy based on AE (AESGS). We will use the MSSAE to implement AESGS, and the specific implementation details of the MSSAE will be provided in Section III-C.

During sample generation, the background dataset $\mathbf{X}_b$ and the target dataset $\mathbf{X}_t$ are processed by the MSSAE to produce diverse and balanced samples. To ensure unique reconstructions in each iteration, a dynamic update strategy is employed. To avoid generating overly similar or low-quality samples, we introduce a screening mechanism based on a loss function threshold. This mechanism ensures that only those reconstructed samples whose loss $\mathcal{L}_e$ is below a preset threshold $\mathcal{U}$ are added to the augmented dataset, thereby ensuring the quality of the samples. In addition, we set an upper limit on the number of samples generated, serving as a termination condition for the algorithm. When the number of generated samples reaches this predefined value, all generated background samples $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_K]$ and $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_K]$ are returned, where $\mathbf{b}_i$ and $\mathbf{t}_j$ represent the spectral vectors of the background samples and target samples, respectively, and K indicates the number of samples.

Through this approach, our method not only effectively increases the number of samples but also ensures that each sample is meaningful and valuable, thereby significantly enhancing the performance and robustness of the model in HSI target detection tasks.

C. Multiscale Spatial–Spectral Autoencoder

In Sections III-A and III-B, we discuss the construction of the training dataset. Here, we design the MSSAE, which enhances feature extraction by overcoming the limitations of single-scale methods. The MSSAE employs an encoder–decoder architecture, where the encoder extracts features from hyperspectral data and the decoder reconstructs them. Both modules incorporate CNN and attention mechanisms to improve feature extraction and reconstruction accuracy.

Let the input data be represented as $\mathbf{X}^{\text{Batch}} \in \mathbb{R}^{B \times 1 \times L}$, where B is the batch size and L is the number of spectral bands. Before feeding the input data into the encoder, it is first mapped into a new space using a linear layer and a 1-D convolution layer, as follows:

$$\mathbf{H}^{(0)} = \text{Covn1d}(\text{Linear}(\mathbf{X}^{\text{Batch}})) \quad (8)$$

where $\mathbf{H}^{(0)} \in \mathbb{R}^{B \times C \times D}$ is the output result after the mapping, C is the mapped number of channels, and D is the mapped length.

In the encoder, to extract features at different scales, we set up three parallel branches for feature extraction, with each branch

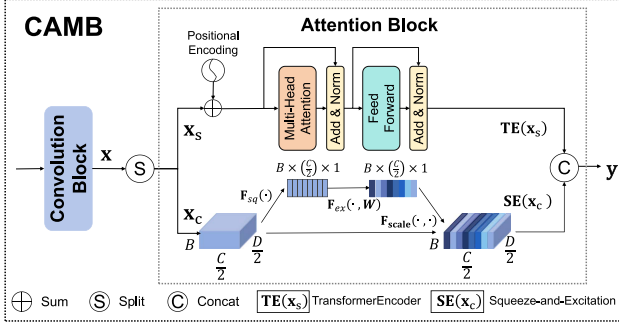


Fig. 2. Flowchart of the CAMB.

using convolution kernels of different sizes (7, 5, and 3, respectively). Meanwhile, to enable the network to simultaneously focus on both local and global information, each branch incorporates a convolution-attention mixed block (CAMB), where each CAMB consists of two components: a Convolution Block (CB) and an Attention Block (AB). Different from generic multiscale methods that perform feature fusion mainly through multikernel convolution and subsequent aggregation, the proposed design embeds attention modeling within each scale-specific branch. In this way, spatial-spectral information is jointly refined at each scale before multiscale fusion, leading to more effective scale-aware feature representation. The flowchart for the CAMB is shown in Fig. 2. The feature extraction process for the j th CAMB in the i th branch is given by

$$\mathbf{H}_i^{(j)} = \text{AB} \left(\text{CB} \left(\mathbf{H}_i^{(j-1)} \right) \right), j = 1, \dots, N. \quad (9)$$

The output of the CB is given by

$$\mathbf{y} = \begin{cases} f(\text{BN}(\text{Conv1d}(\mathbf{x}))), & \text{stride} = 2 \\ \delta_D(f(\text{BN}(\text{Conv1d}(\mathbf{x}))), & \text{stride} = 1 \end{cases} \quad (10)$$

where $\mathbf{y} \in \mathbb{R}^{B \times C \times (\frac{D}{2})}$ is the output of the CB, BN is a 1-D batch normalization layer, $f(\cdot)$ is the ReLU activation function, and $\delta_D(\cdot)$ is a 1-D max pooling layer with a pooling factor of 2. It is worth noting that, for convolution operations with a kernel size of 3, directly reducing the feature map size by using a stride of 2 may result in the loss of important details. To address this issue, we set the stride of the 3×3 convolution to 1 and apply a MaxPool operation after the convolution to achieve feature map downsampling.

The AB consists of two key components: a channel attention mechanism and a spatial-spectral attention mechanism. These are implemented using the Squeeze-and-Excitation (SE) block [44] and the TransformerEncoder (TE) block [45], respectively. Let the input data to the AB be denoted as $\mathbf{x} \in \mathbb{R}^{B \times C \times (\frac{D}{2})}$. We first split the input along the channel dimension into two parts, $\mathbf{x}_c \in \mathbb{R}^{B \times (\frac{C}{2}) \times (\frac{D}{2})}$ and $\mathbf{x}_s \in \mathbb{R}^{B \times (\frac{C}{2}) \times (\frac{D}{2})}$, which are then fed into the SE and TE blocks, respectively. This process can be expressed as follows:

$$\mathbf{y} = \text{concat}([\text{SE}(\mathbf{x}_c), \text{TE}(\mathbf{x}_s)], \text{dim}=1) \quad (11)$$

where $\mathbf{y} \in \mathbb{R}^{B \times C \times (\frac{D}{2})}$ is the output of AB.

The output from each branch, denoted as $\mathbf{H}_i^{(N)} \in \mathbb{R}^{B \times C \times (\frac{D}{2^N})}$, is obtained through the above calculations, where N is the number of CAMBs in each branch. The final fused result $\mathbf{Y}^{(0)}$ is then computed as follows:

$$\mathbf{Y}^{(0)} = \sum_{i=1}^3 \mathbf{H}_i^{(N)}. \quad (12)$$

Finally, a 1-D convolution layer is employed to compress and map the features, producing the final encoded result. Specifically, this process can be described by the following equation:

$$\mathbf{O}_e^{\text{Batch}} = \text{reshape} \left(\text{Conv1d} \left(\mathbf{Y}^{(0)} \right) \right) \quad (13)$$

where $\mathbf{O}_e^{\text{Batch}} \in \mathbb{R}^{B \times L_e}$ is the result of encoding the input data, with a dimension of L_e .

To reconstruct the input data, the decoder performs the inverse mapping of the encoded features using a single-branch structure. Each layer consists of upsampling (deconvolution) and an AB. The output of the j th layer, $\mathbf{Y}^{(j)}$, is calculated as

$$\mathbf{Y}^{(j)} = \text{AB} \left(\text{CB} \left(\delta_U \left(\mathbf{Y}^{(j-1)} \right) \right) \right), j = 1, \dots, N \quad (14)$$

where $\delta_U(\cdot)$ is the upsampling function. In this article, the upsampling function adopts the nearest-neighbor interpolation mode, with a scaling factor set to 2.

After performing layerwise deconvolutions on the features, the final reconstructed output is obtained by

$$\mathbf{O}_d^{\text{Batch}} = \text{Linear} \left(\text{Conv1d} \left(\mathbf{Y}^{(N)} \right) \right). \quad (15)$$

D. Clustered Adaptive Focus Training Strategy

In this section, we will detail how the proposed network learns from the training samples. Due to spectral distortion and spectral mixing during the acquisition of the HSI, the spectral features of background and target pixels can be very similar, which negatively impacts the accuracy of target detection. If the SAM algorithm is directly applied for detection, these similar pixels may lead to severe detection anomalies. To address this issue, we propose the CAFTS, which synergistically optimizes the feature space through clustered sampling and AEW-Loss. Specifically, by partitioning samples into similarity-based clusters and sampling uniformly from each cluster, the strategy naturally includes easy, moderate, and hard samples in each epoch, thereby preserving the overall data distribution. AEW-Loss further assigns smoothly varying weights based on normalized similarity, ensuring that easy samples are not ignored, while more ambiguous ones receive additional focus. Together, these mechanisms enable the MSSAE to learn discriminative features effectively without overfitting to extreme outliers.

1) *Clustered Sampling*: As previously mentioned, we denote the enhanced background and target training datasets as $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_K]$ and $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_K]$, where $\mathbf{b}_i$ and $\mathbf{t}_j$ represent the spectral vectors of the background and target samples, respectively, and K is the number of background and target samples. We cluster the background and target datasets into multiple categories, and the basic procedure is as follows.

First, calculate the similarity of each sample in $\mathbf{B}$ and $\mathbf{T}$ to the prior target vector $\mathbf{t}$. The similarities are computed according to the following formula:

$$\mathbf{S}_b[i] = \text{SAS}(\mathbf{B}[i], \mathbf{t}), \mathbf{S}_t[j] = \text{SAS}(\mathbf{T}[j], \mathbf{t}) \quad (16)$$

where $\mathbf{S}_b$ is the similarity vector between the background sample matrix $\mathbf{B}$ and the prior target vector $\mathbf{t}$, and $\mathbf{S}_t$ is the similarity vector between the target sample matrix $\mathbf{T}$ and the prior target vector $\mathbf{t}$.

Next, we rank the background and target samples based on the similarity vectors $\mathbf{S}_b$ and $\mathbf{S}_t$. Since we are concerned with the internal ordering of the samples, we individually sort the background sample matrix $\mathbf{B}$ and the target sample matrix $\mathbf{T}$. Here, we employ a sorting function $\text{Sort}(\mathbf{X}, \mathbf{S})$ to reorder the sample matrix $\mathbf{X}$ based on the similarity vector $\mathbf{S}$, calculated as follows:

$$\mathbf{B}' = \text{Sort}(\mathbf{B}, \mathbf{S}_b), \mathbf{T}' = \text{Sort}(\mathbf{T}, \mathbf{S}_t) \quad (17)$$

where $\mathbf{B}'$ and $\mathbf{T}'$ represent the sorted background and target sample matrices, respectively, with the sorting being done in the descending order of similarity. $\mathbf{B}'$ and $\mathbf{T}'$ are then evenly divided into n category sets, resulting in $\mathcal{C}_{B'} = \{\mathbf{B}'_1, \mathbf{B}'_2, \dots, \mathbf{B}'_n\}$ and $\mathcal{C}_{T'} = \{\mathbf{T}'_1, \mathbf{T}'_2, \dots, \mathbf{T}'_n\}$.

Finally, initialize a training set $\mathcal{D}_{\text{train}} = \emptyset$. In each iteration, we randomly select b samples from each category set of the background and target samples to form a batch of data, which is then added to $\mathcal{D}_{\text{train}}$. This process continues until all samples in the category sets have been selected. Ultimately, the training set $\mathcal{D}_{\text{train}}$ consists of all the batched spatial-spectral data $\mathbf{X}^{\text{Batch}}$.

2) *Loss Function*: In the model optimization section, we introduce an AEW-Loss function as the loss function for the encoder. By calculating the spectral angle similarity and performing adaptive adjustments, AEW-Loss ensures that the encoder can effectively capture the key information in the input data. During the computation of the encoder loss, we first calculate the similarity coefficients between the batched spectral data $\mathbf{X}^{\text{Batch}}$ and the prior target vector $\mathbf{t}$. In order to improve the ability of the model to separate targets from the background, we normalize the similarity coefficients using the following equation:

$$C[i] = \alpha + \text{Norm}(\text{SAS}(\mathbf{X}^{\text{Batch}}[i], \mathbf{t}), \dim = 0) \quad (18)$$

where $\text{Norm}(\cdot, \dim = 0)$ is the standardization operation performed on the data along the 0th dimension, and in this article, the max-min normalization method is used. α is a dynamically adjusted bias term that helps prevent numerical instability and fine-tunes the contribution ratio of the loss across different samples.

Next, we compute the similarity between the encoded batch spectral data $\mathbf{O}_e^{\text{Batch}}$ and the encoded prior target vector $\mathbf{O}_e^t$. The calculation of the i th element in the similarity vector $\mathbf{S}$ is given by the following equation:

$$\mathbf{S}[i] = \text{SAS}(\mathbf{O}_e^{\text{Batch}}[i], \mathbf{O}_e^t). \quad (19)$$

Combining the above calculations, the final computation of the encoder loss L_e is given by the following equation:

$$L_e = \frac{1}{B} \cdot (L_e^t + L_e^b) \quad (20)$$

$$L_e^t = \sum_{i \in \text{id}_t} \exp\left(\frac{1}{C[i]}\right) \cdot (1 - \mathbf{S}[i]) \quad (21)$$

$$L_e^b = \sum_{j \in \text{id}_b} \exp(C[j]) \cdot \mathbf{S}[j] \quad (22)$$

where id_t and id_b represent the index positions of the target and background samples within the batch data, respectively. The encoder loss is the weighted sum of the positive and negative sample losses. Through this loss function, the model is ensured to effectively distinguish between different spectral features

$$L_d = \frac{1}{B} \cdot \left(\sum_{i=1}^B (\varepsilon \cdot L_{sa} + L_{mse}) \right) \quad (23)$$

$$L_{sa} = \text{SA}(\mathbf{O}_d^{\text{Batch}}[i], \mathbf{X}^{\text{Batch}}[i]) \quad (24)$$

$$L_{mse} = \text{MSE}(\mathbf{O}_d^{\text{Batch}}[i], \mathbf{X}^{\text{Batch}}[i]) \quad (25)$$

where ε is the balancing coefficient between the spectral angle loss and the mean squared error loss, and it is consistently set to 0.9 in all experiments conducted in this article.

By integrating the encoder loss and the decoder loss, the total loss is calculated as follows:

$$L = \lambda \cdot L_e + L_d \quad (26)$$

where λ is the balancing coefficient between the encoder loss and the decoder loss, and it is consistently set to 0.8 in all experiments conducted in this article.

E. Target Detection

During the testing phase, the constructed HSI data $\mathbf{X}^{\text{avg}}$ are used as the test input. Each SV in $\mathbf{X}^{\text{avg}}$, together with the prior target spectral vector $\mathbf{t}$, is fed into the pretrained MSSAE encoder to obtain feature representations, after which target detection is performed based on spectral angle similarity.

Let O_e^i and O_e^t represent the results of encoding the i th SV $\mathbf{x}_i$ and the prior target spectral vector $\mathbf{t}$, respectively. The calculation of the similarity score is given by the following equation:

$$S_i = 1 - \frac{\cos^{-1}\left(\frac{\mathbf{x}_i^T \cdot \mathbf{t}}{\|\mathbf{x}_i\| \cdot \|\mathbf{t}\|}\right)}{\pi} \quad (27)$$

where S_i represents the similarity score between $\mathbf{x}_i$ and $\mathbf{t}$. The final detection result, $\mathbf{S} = [S_1, S_2, \dots, S_{H \times W}]$, is obtained by calculating the similarity scores between all spectral vectors and the prior target spectral vector.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Hyperspectral Datasets

To evaluate the effectiveness of the proposed method, experiments are conducted on five publicly available HSI datasets covering diverse scenarios, including SanDiego-1, Airport-2, Beach-2, Urban-2, and Qingpu-1.

1) *SanDiego Dataset*: The first dataset, SanDiego-1, is collected by the Airborne Visible/Infrared Imaging Spectrometer

on 9 November 2011 at San Diego International Airport in California, USA [46]. It has a spatial resolution of 3.5 m and retains 189 spectral bands after removing low-quality bands (such as water vapor absorption bands and bands with low signal-to-noise ratio) [47]. This dataset consists of 120×120 pixels, with the aircraft targets of interest occupying 64 pixels.

2) *Airport-Beach-Urban (ABU) Dataset*: The second to fourth datasets are derived from the ABU dataset¹ [48]. The second dataset, Airport-2, has a spatial resolution of 7.1 m and contains 205 spectral bands. It comprises 100×100 pixels, with the aircraft targets of interest occupying 87 pixels. The third dataset, Beach-2, has a spatial resolution of 7.5 m and contains 193 spectral bands. It also comprises 100×100 pixels, with the targets of interest occupying 202 pixels. The fourth dataset, Urban-2, has a spatial resolution of 17.2 m and contains 207 spectral bands. It comprises 100×100 pixels, with the building targets of interest occupying 155 pixels.

3) *Qingpu Dataset*: The fifth dataset, Qingpu-1² [49], is acquired by the Airborne Multimodal Imaging Spectrometer (AMMIS) over Qingpu District, Shanghai, China. It has a spatial resolution of 0.75 m and covers the wavelength range from 400 to 1000 nm, consisting of 250 spectral bands. After atmospheric correction, radiometric calibration, and image cropping of the original strip data, an HTD dataset is obtained. In our experiments, the image is cropped to a size of 150×150 pixels, where three vehicles are considered as targets, corresponding to a total of 31 target pixels.

B. Evaluation Indicators

To comprehensively assess target detection in HSIs, we employ multiple evaluation methods, including detection map visualization, 3-D receiver operating characteristic (3D-ROC) analysis [50], and background-target separation visualization.

In 3D-ROC analysis, target and background pixels are distinguished by adjusting the detection threshold τ . True positives (TP) and false negatives (FN) correspond to target pixels above and below τ , respectively, while false positives (FP) and true negatives (TN) apply similarly to background pixels. For 2-D ROC curves, three key curves are derived— (P_f, P_d) , (τ, P_f) , and (τ, P_d) , which measure overall detection performance, background suppression effect, and target detection performance, respectively. The true positive rate (P_d) and false alarm rate (P_f) are computed as

$$P_d = \frac{TP}{TP + FN}, \quad P_f = \frac{FP}{TN + FP}. \quad (28)$$

We calculate the area under the 2-D ROC curves, denoted as $AUC(P_d, P_f)$, $AUC(\tau, P_d)$, and $AUC(\tau, P_f)$. Higher values for the first two indicate better detection performance, while a lower value for $AUC(\tau, P_f)$ is preferable. In addition, two composite metrics [51] provide a more comprehensive assessment

$$AUC_{OA} = AUC(P_d, P_f) + AUC(\tau, P_d) - AUC(\tau, P_f) \quad (29)$$

$$AUC_{SNPR} = \frac{AUC(\tau, P_d)}{AUC(\tau, P_f)}. \quad (30)$$

Box plots illustrate the central tendency and spread of detection scores, with a more compact box indicating a concentrated data distribution.

C. Experimental Setup for the Proposed SBSNet

In the experiment of this article, the proposed MSSAE involves several key hyperparameter settings. First, in the mapping layer, the channel dimension of the input data is mapped to 32, and the spectral dimension is mapped to 128. Each branch in the network contains two CAMB blocks, where each CAMB includes one TE block with a depth of 1, a hidden layer size of 128, and two attention heads.

In the training data preparation phase, we utilize prior target spectra to construct target and background pixel sets from the unlabeled HSI and then apply the AESGS to augment them. The amount of data for all experiments are augmented to ten times the number of raw HSI pixels, ensuring that the data for both target and background are approximately equal. Small batches of size $64 \times 10 \times 2$ are then constructed as input, where 10×2 indicates that the background and target samples are divided into ten categories each, and 64 samples are drawn from each category to form a batch. During the testing phase, the testing data consist of the raw unlabeled HSI.

For the loss function, the balance coefficients ε and λ in the decoder loss and total loss are fixed at 0.9 and 0.8, respectively. To optimize the training of the MSSAE, we utilize the Adam optimizer with an initial learning rate of $1e-4$. The learning rate decays ten times throughout the training, with each decay reducing the rate to 75% of the previous value. The number of epochs is set to 200 for all five datasets.

In addition, two important hyperparameters need to be determined: the window size S in the LSFFM and the bias term α in AEW-Loss. In all experiments of this article, we set the window size S and the bias term α to (5, 0.8), (5, 0.8), (3, 0.5), (3, 0.2), and (3, 0.8) for the SanDiego-1, Airport-2, Beach-2, Urban-2, and Qingpu-1 datasets, respectively.

Finally, all experiments are implemented in a Python 3.9 and PyTorch 2.3.0 environment and are executed on an NVIDIA 3090 GPU.

D. Comparison Methods

To illustrate the effectiveness and superiority, we compare our proposed SBSNet with ten different detectors—SAM³ [30], CEM³ [13], OSP⁴ [14], E-CEM³ [34], BLTSC⁵ [38], HTD-IRN⁶ [41], OS-VAE⁷ [40], DPN-CSSTD⁸ [29], HTD-DMCG [52], and HTD-Mamba [53]—from both an objective

³[Online]. Available: https://github.com/ruizhaocv/E_CEM-for-Hyperspectral-Target-Detection.git

⁴[Online]. Available: <https://github.com/safakgunes/Hyperspectral-Images-Target-Detection.git>

⁵[Online]. Available: <https://github.com/zhangxin-xd/BLTSC.git>

⁶[Online]. Available: <https://github.com/shendb2022/HTD-IRN.git>

⁷[Online]. Available: <https://github.com/CX-He/OS-VAE.git>

⁸[Online]. Available: <https://github.com/Chenxi-29/DPN-CSSTD.git>

¹[Online]. Available: <https://xudongkang.weebly.com/data-sets.html>

²[Online]. Available: <https://www.3sobs.cn/share>

TABLE I
PARAMETER ANALYSIS OF CONVOLUTION KERNEL SIZE COMBINATIONS, EVALUATED BY $AUC_{(P_d, P_f)}$ $\uparrow$, PARAMETERS (M) $\downarrow$, AND FLOPS (G) $\downarrow$

Kernel Set	SanDiego-1			Airport-2			Beach-2			Urban-2			Qingpu-1		
	AUC	Params	FLOPs	AUC	Params	FLOPs	AUC	Params	FLOPs	AUC	Params	FLOPs	AUC	Params	FLOPs
3, 5, 7	0.9997	0.0896	0.0034	0.9947	0.0917	0.0034	0.9971	0.0901	0.0034	0.9991	0.0919	0.0034	0.9999	0.0947	0.0034
3, 5, 9	0.9996	0.0937	0.0036	0.9672	0.0958	0.0036	0.9970	0.0942	0.0036	0.9991	0.0960	0.0036	0.9998	0.1015	0.0036
3, 5, 11	0.9997	0.0978	0.0038	0.9679	0.0999	0.0038	0.9965	0.0983	0.0038	0.9951	0.1001	0.0038	0.9999	0.1056	0.0038
3, 7, 9	0.9996	0.0978	0.0038	0.8595	0.0999	0.0038	0.9970	0.0983	0.0038	0.9965	0.1001	0.0038	0.9999	0.1056	0.0038
3, 7, 11	0.9997	0.1019	0.0040	0.9874	0.1040	0.0040	0.9889	0.1024	0.0040	0.9989	0.1042	0.0040	0.9997	0.1097	0.0040

Bold Denotes the best.

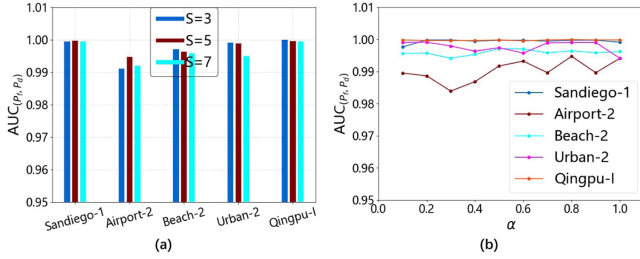


Fig. 3. Sensitivity analysis of key hyperparameters in the SBSNet. (a) Window size S in the LSFFM. (b) Bias term α in AEW-Loss. The SBSNet shows stable performance across different hyperparameter settings.

quantitative analysis and a subjective visual evaluation perspective to demonstrate the superiority of the SBSNet in HTD. The code for all the comparison detectors is obtained from the public repositories of the authors, and the hyperparameters are fine-tuned based on the guidelines provided in their respective papers. In the experiments, the grayscale values of all datasets are normalized to the range $[0,1]$, and for all traditional detectors, the average spectrum after removing spectral bias is chosen as the prior target spectrum.

E. Hyperparameter Sensitivity Analysis

In this section, we analyze the sensitivity of key hyperparameters in the proposed SBSNet, including the convolution kernel sizes in the multiscale feature extraction, the window size S in the LSFFM, and the bias term α in AEW-Loss. This analysis is conducted on five representative HSI datasets: SanDiego-I, Airport-2, Beach-2, Urban-2, and Qingpu-1.

1) *Analysis of Convolution Kernel Sizes:* To investigate the effect of convolution kernel size selection for multiscale feature extraction, we test multiple combinations of kernel sizes across the five datasets. Table I summarizes the detection performance (AUC), number of parameters, and FLOPs for different kernel sets. As observed, the combination of (3,5,7) achieves consistently high detection accuracy while maintaining relatively fewer parameters and lower computational cost compared to other configurations. This kernel configuration provides an effective balance between fine-grained local feature modeling and larger receptive fields, capturing both fine-grained and contextual spectral-spatial information, which is crucial for HTD where targets can vary in size and spatial distribution.

2) *Analysis of Window Size S in the LSFFM:* We evaluate three different window sizes for the LSFFM, namely 3, 5, and 7, across the five datasets. As shown in Fig. 3(a), varying the

window size results in only minor fluctuations in detection performance, indicating that the SBSNet is stable with respect to this hyperparameter. For datasets with larger or more spatially dispersed targets, such as SanDiego-I and Airport-2, the best performance is achieved when the window size is set to 5. By contrast, datasets characterized by smaller or more spatially concentrated targets, including Beach-2, Urban-2, and Qingpu-1, achieve higher performance when a smaller window size of 3 is used.

3) *Analysis of Bias Term α in AEW-Loss:* The bias term in AEW-Loss is evaluated over the range from 0.1 to 1 with an increment of 0.1. The results show that the optimal bias term varies slightly across datasets: SanDiego-I and Airport-2 achieve their best performance when the bias term is set to 0.8, Beach-2 performs best at 0.5, Urban-2 favors a smaller value of 0.2, and Qingpu-1 again reaches optimal performance at 0.8. As shown in Fig. 3(b), adjusting the bias term leads to only minor performance variations, indicating that the SBSNet remains relatively robust with respect to this hyperparameter.

F. Analysis and Comparison of Experimental Results

1) *Quantitative Analysis of Experimental Results:* We quantitatively evaluate the detection performance of the SBSNet by comparing it with representative HTD methods under multiple AUC-based metrics. The comprehensive detection results across five datasets are summarized in Table II, while the comparison in terms of model complexity and computational efficiency is reported in Table III. The following analysis focuses on overall detection consistency, performance tradeoffs among different methods, and robustness across diverse scenes.

As shown in Table II, the SBSNet demonstrates consistently strong detection performance across all datasets under multiple AUC-based evaluation metrics. In particular, the SBSNet achieves the best or joint-best results on $AUC_{(P_d, P_f)}$ and AUC_{OA} in most cases, indicating that it maintains a high target detection probability while effectively controlling false alarms over a wide range of thresholds. This consistent behavior across diverse scenes suggests that the SBSNet exhibits strong robustness to variations in target size, background complexity, and spatial distribution. It is worth noting that on certain datasets and specific metrics, such as $AUC_{(\tau, P_d)}$ on Urban-2, some competing methods (e.g., HTD-DMCG) may achieve higher target enhancement scores. However, these gains are often accompanied by inferior background suppression performance, as reflected by higher $AUC_{(\tau, P_f)}$ values. In contrast, the SBSNet maintains a more balanced performance profile, achieving

TABLE II
PERFORMANCE COMPARISON OF COMPETING HTD METHODS IN TERMS OF AUC SCORES ON FIVE DATASETS

Datasets	Metrics	Classical HTD Methods				DL-based HTD Methods						
		SAM	CEM	OSP	E-CEM	BLTSC	HTD-IRN	OS-VAE	DPN-CSSTD	HTD-DMCG	HTD-Mamba	SBSNet
SanDiego-1	$AUC_{(P_d, P_f)} \uparrow$	0.9956	0.9924	0.9852	0.9476	0.9997	0.9990	0.9988	0.9972	0.9970	0.9990	0.9997
	$AUC_{(\tau, P_d)} \uparrow$	0.3459	0.7258	0.5272	0.6340	0.5766	0.5327	0.0316	0.9540	0.9135	0.7344	0.9718
	$AUC_{(\tau, P_f)} \downarrow$	0.0227	0.4208	0.0910	0.1864	0.0038	<u>0.0029</u>	0.0009	0.0780	0.0108	0.0053	0.0081
	$AUC_{OA} \uparrow$	1.3188	1.2974	1.4213	1.3951	1.5726	1.5287	1.0294	1.8731	<u>1.8997</u>	1.7281	1.9634
	$AUC_{SNPR} \uparrow$	15.2307	1.7249	5.7928	3.4008	<u>152.1478</u>	181.8055	36.2644	12.2302	84.5120	138.3433	120.2698
Airport-2	$AUC_{(P_d, P_f)} \uparrow$	0.7231	0.8605	0.8130	0.9603	0.9795	0.7592	0.9901	0.9871	0.8207	<u>0.9927</u>	0.9947
	$AUC_{(\tau, P_d)} \uparrow$	0.3202	0.3856	0.3699	0.5539	0.3218	0.1415	0.0379	0.0344	0.5240	<u>0.5403</u>	0.6512
	$AUC_{(\tau, P_f)} \downarrow$	0.1548	0.1567	0.1588	0.1032	<u>0.0016</u>	0.0488	0.0013	0.0154	0.0237	0.0081	0.0032
	$AUC_{OA} \uparrow$	0.8886	1.0893	1.0241	1.4110	1.2997	0.8519	1.0267	1.0061	1.3210	<u>1.5249</u>	1.6427
	$AUC_{SNPR} \uparrow$	2.0685	2.4605	2.3295	5.3680	<u>197.4417</u>	2.9000	28.9542	2.2275	22.1190	66.6190	206.7238
Beach-2	$AUC_{(P_d, P_f)} \uparrow$	0.9472	<u>0.9938</u>	<u>0.9938</u>	0.9537	0.9924	0.9747	0.9922	0.9786	0.7588	0.9904	0.9971
	$AUC_{(\tau, P_d)} \uparrow$	0.1977	0.6168	0.5092	0.5595	0.5462	0.7310	0.0496	0.8194	0.6814	0.6582	0.9882
	$AUC_{(\tau, P_f)} \downarrow$	0.0536	0.2167	0.0336	0.1294	<u>0.0087</u>	0.2578	0.0042	0.2609	0.0879	0.0145	0.0106
	$AUC_{OA} \uparrow$	1.0914	1.3939	1.4694	1.3838	1.5299	1.4479	1.0375	1.5371	1.3523	<u>1.6341</u>	1.8848
	$AUC_{SNPR} \uparrow$	3.6904	2.8459	15.1424	4.3247	<u>62.8585</u>	2.8351	11.7281	3.1410	7.7540	45.3430	85.1374
Urban-2	$AUC_{(P_d, P_f)} \uparrow$	0.9255	0.8835	0.9420	0.9752	0.9864	0.9982	0.9191	0.9971	0.9990	0.9462	0.9991
	$AUC_{(\tau, P_d)} \uparrow$	0.2090	<u>0.7661</u>	0.1713	0.5568	0.4437	0.3910	0.0302	0.1852	0.9955	0.3855	0.7624
	$AUC_{(\tau, P_f)} \downarrow$	0.0296	0.6120	0.0606	0.0963	0.0023	0.0307	<u>0.0013</u>	0.0215	0.0124	0.0230	0.0009
	$AUC_{OA} \uparrow$	1.1050	1.0376	1.0524	1.4356	1.4278	1.3585	0.9480	1.1608	1.9821	1.3086	<u>1.7606</u>
	$AUC_{SNPR} \uparrow$	7.0687	1.2518	2.8273	5.7805	<u>193.7686</u>	12.7262	23.5938	8.6326	80.2072	16.7267	837.8462
Qingpu-1	$AUC_{(P_d, P_f)} \uparrow$	0.9932	0.9816	0.9807	0.9678	0.9994	0.9997	0.8728	0.9999	0.9999	0.9998	0.9999
	$AUC_{(\tau, P_d)} \uparrow$	0.3964	0.6776	0.4843	0.6435	0.2850	0.6536	0.0071	1.0000	1.0000	0.8741	<u>0.9973</u>
	$AUC_{(\tau, P_f)} \downarrow$	0.0166	0.3033	0.1001	0.0610	0.0010	0.0622	0.0025	0.0034	0.0015	0.0098	0.0025
	$AUC_{OA} \uparrow$	1.3730	1.3558	1.3649	1.5503	1.2835	1.5911	0.8775	<u>1.9966</u>	1.9985	1.8640	1.9947
	$AUC_{SNPR} \uparrow$	23.9228	2.2338	4.8387	10.5518	293.8041	10.5126	2.8898	296.7359	657.8882	88.9217	<u>395.7421</u>

Bold denotes the best and Underlined denotes the second-best.

TABLE III
PERFORMANCE COMPARISON OF COMPETING DL-BASED HTD METHODS IN TERMS OF $AUC_{(P_d, P_f)}$ AND EFFICIENCY METRICS, INCLUDING PARAMETERS (M), FLOPS (G), AND TESTING TIME (S), ON FIVE DATASETS

Datasets	Metrics	BLTSC	HTD-IRN	OS-VAE	DPN-CSSTD	HTD-DMCG	HTD-Mamba	SBSNet
SanDiego-1	$AUC_{(P_d, P_f)} \uparrow$	0.9997	0.9990	0.9988	0.9972	0.9970	0.9990	0.9997
	Params $\downarrow$	1.5660	0.4090	1.0012	0.5079	<u>0.2002</u>	0.3252	0.0896
	FLOPs $\downarrow$	1.8954	0.0036	0.0020	<u>0.0018</u>	0.0044	0.0017	0.0034
	Testing Time $\downarrow$	8.130	0.970	3.348	<u>1.046</u>	2.327	3.754	2.487
Airport-2	$AUC_{(P_d, P_f)} \uparrow$	0.9795	0.7592	0.9901	0.9871	0.8207	<u>0.9927</u>	0.9947
	Params $\downarrow$	1.5789	0.4112	1.0173	0.5497	<u>0.2002</u>	0.5071	0.0917
	FLOPs $\downarrow$	1.9115	0.0036	0.0020	0.0020	0.0046	0.0159	<u>0.0034</u>
	Testing Time $\downarrow$	7.773	<u>0.990</u>	2.975	0.860	1.658	3.597	1.789
Beach-2	$AUC_{(P_d, P_f)} \uparrow$	0.9924	0.9747	0.9922	0.9786	0.7588	0.9904	0.9971
	Params $\downarrow$	1.5692	0.4096	1.0053	0.5183	<u>0.2002</u>	0.4948	0.0901
	FLOPs $\downarrow$	1.8994	0.0036	<u>0.0020</u>	0.0018	0.0044	0.0148	0.0034
	Testing Time $\downarrow$	7.690	<u>0.980</u>	2.760	0.847	2.099	3.771	2.007
Urban-2	$AUC_{(P_d, P_f)} \uparrow$	0.9864	0.9982	0.9191	0.9971	<u>0.9990</u>	0.9462	0.9991
	Params $\downarrow$	1.5805	0.4115	1.0193	0.5550	<u>0.2002</u>	0.5092	0.0919
	FLOPs $\downarrow$	1.9136	0.0036	0.0020	0.0020	0.0048	0.0159	<u>0.0034</u>
	Testing Time $\downarrow$	7.713	<u>0.990</u>	2.697	0.887	2.153	3.597	1.941
Qingpu-1	$AUC_{(P_d, P_f)} \uparrow$	0.9994	0.9997	0.8728	0.9999	0.9999	<u>0.9998</u>	0.9999
	Params $\downarrow$	1.6149	0.4174	1.0623	0.5550	<u>0.2002</u>	0.5532	0.0974
	FLOPs $\downarrow$	1.9579	0.0037	<u>0.0021</u>	0.0020	0.0050	0.0192	0.0034
	Testing Time $\downarrow$	9.108	<u>1.533</u>	3.632	0.887	4.393	5.328	4.396

Bold denotes the best and underlined denotes the second-best.

competitive target enhancement while simultaneously suppressing background responses more effectively. This balance is further supported by its superior AUC_{SNPR} scores on most datasets, which quantify the overall separation between target and background. These results indicate that the SBSNet does not optimize a single metric in isolation but instead provides a well-rounded detection capability.

To further analyze the tradeoff between detection accuracy and computational efficiency, Table III focuses on a detailed comparison among DL-based HTD methods. Compared with existing approaches, the SBSNet achieves a favorable balance between performance and efficiency. While some lightweight models, such as HTD-IRN, exhibit shorter testing times, their detection accuracy, especially $AUC_{(P_d, P_f)}$, is noticeably lower

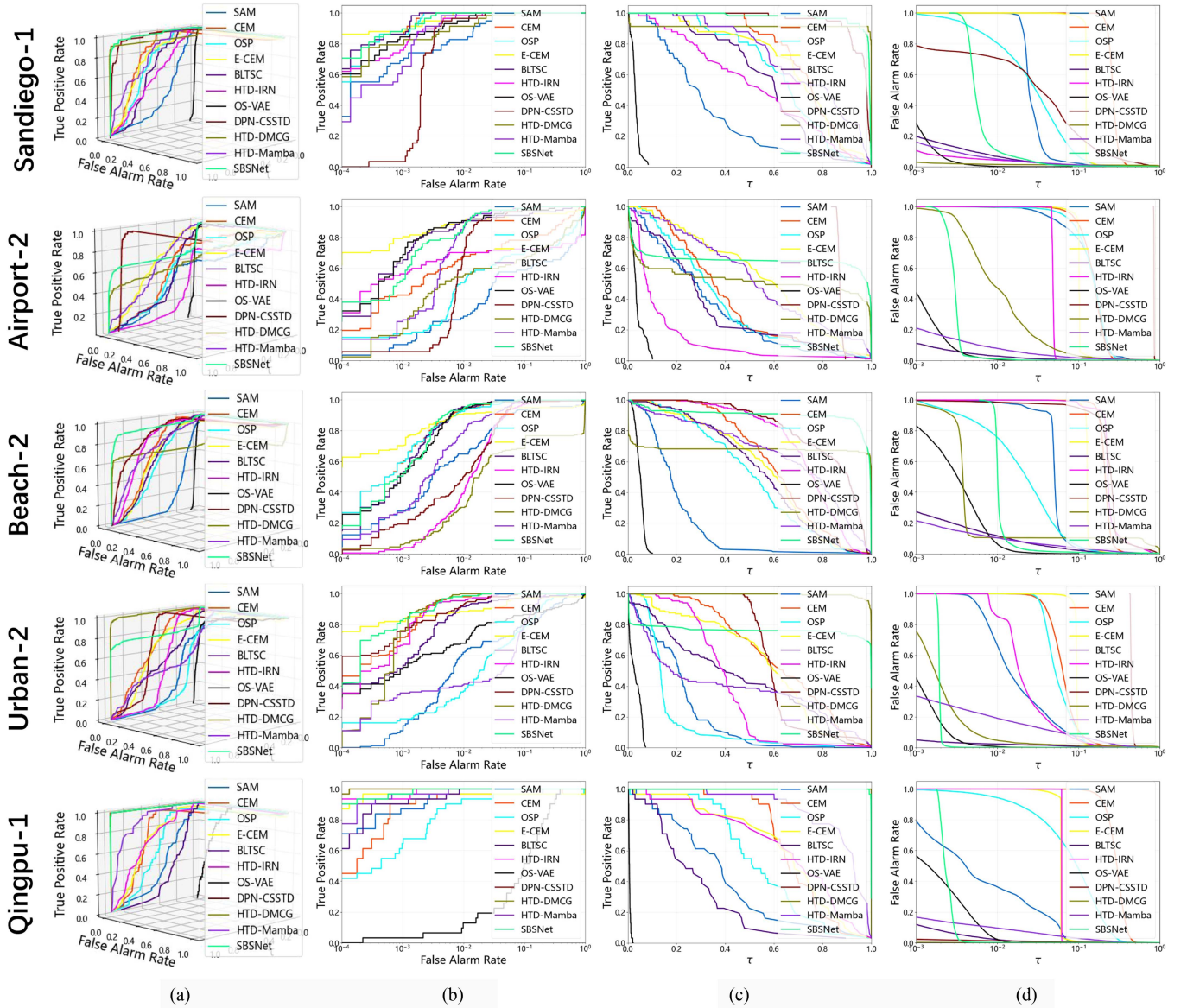


Fig. 4. ROC curves of the SBSNet and ten competing detectors on five benchmark datasets. (a) 3-D ROC curves. (b) 2-D ROC (P_f, P_d). (c) 2-D ROC (τ, P_d). (d) 2-D ROC (τ, P_f).

than that of the SBSNet across most datasets. This indicates that efficiency gains in these methods may come at the cost of reduced detection reliability. On the other hand, methods such as HTD-DMCG and HTD-Mamba demonstrate strong performance on specific datasets or metrics, but their parameter sizes, FLOPs, or performance stability across datasets reveal certain limitations. For example, HTD-DMCG achieves outstanding results on Urban-2, yet its performance degrades on more challenging scenes such as Beach-2, suggesting potential sensitivity to data characteristics. In comparison, the SBSNet maintains competitive or superior detection accuracy across all datasets while using significantly fewer parameters and moderate computational complexity. These observations indicate that the SBSNet provides a more practical tradeoff among accuracy, efficiency, and model compactness, which is particularly important for large-scale or resource-constrained HTD applications.

The quantitative results are further supported by the visual comparisons in Figs. 4 and 5. As shown in the 3-D ROC curves and the 2-D ROC (P_f, P_d) plots, the SBSNet consistently achieves larger enclosed areas than competing methods, indicating a superior tradeoff between detection probability and false alarm rate. Meanwhile, the background–target separation plots reveal that the SBSNet produces more compact and well-separated distributions for target and background responses, indicating stable and reliable detection behavior. These visualizations provide intuitive evidence that complements the quantitative results and further substantiates the effectiveness of the proposed SBSNet.

2) *Visual Evaluation of Experimental Results:* The visual detection results of the SBSNet and comparative methods across five datasets are shown in Fig. 6. For SAM, CEM, and OSP, good background suppression is observed in the SanDiego-1,

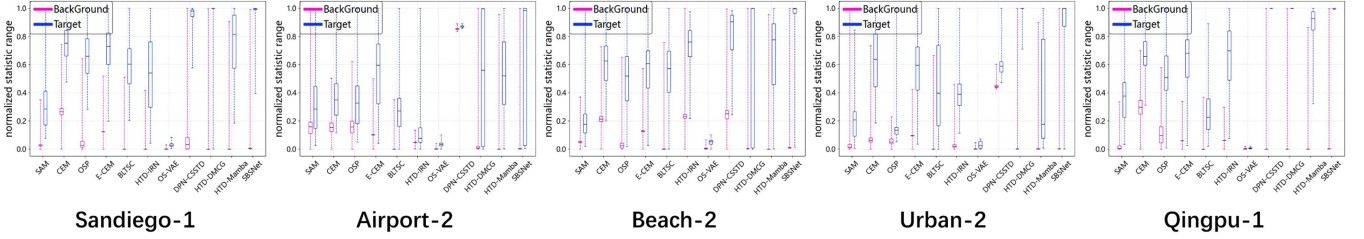


Fig. 5. Box plots of the SBSNet and ten different detectors on five benchmark datasets.

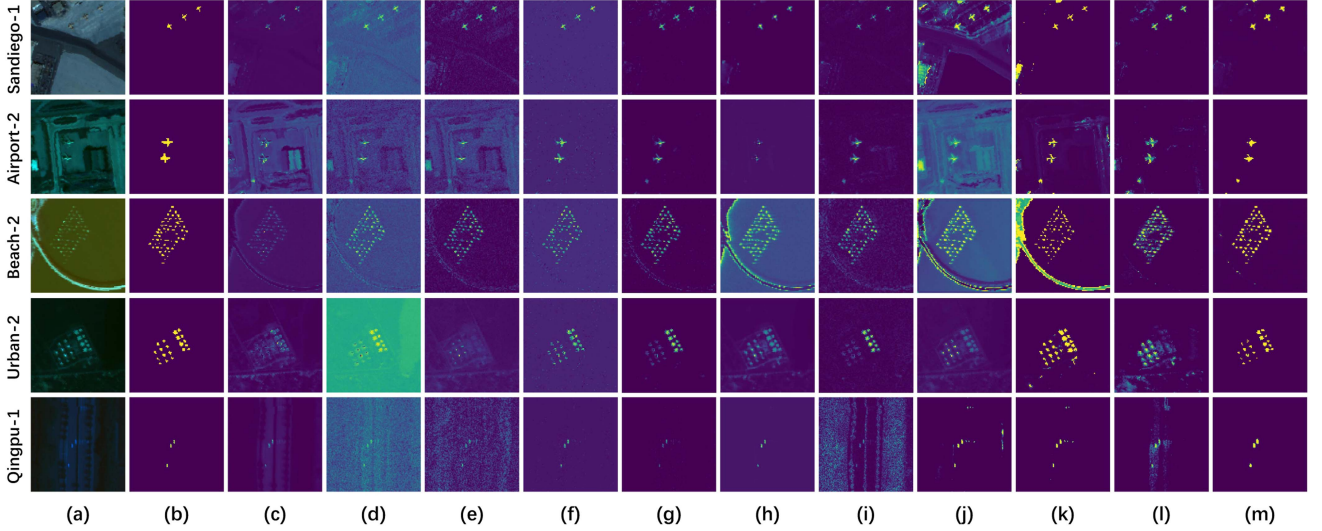


Fig. 6. Comparison of detection results between SBSNet and ten competing detectors on five benchmark datasets. (a) Pseudocolor image. (b) Ground truth. (c) SAM. (d) CEM. (e) OSP. (f) E-CEM. (g) BLTSC. (h) HTD-IRN. (i) OS-VAE. (j) DPN-CSSTD. (k) HTD-DMCG. (l) HTD-Mamba. (m) SBSNet (ours).

Beach-2, Urban-2, and Qingpu-1 datasets. However, in the more complex background of the Airport-2 dataset, these methods show poor target detection performance, with targets easily obscured by the background. E-CEM provides relatively stable detection results, yet its target enhancement capability remains limited, leading to low target-background contrast. Among the DL-based methods, HTD-IRN achieves reasonable background suppression but fails to consistently highlight targets in complex scenes. BLTSC and OS-VAE generate visually clear detection maps but suffer from insufficient separation between target and background responses. DPN-CSSTD tends to produce strong target activations but also introduces considerable background interference. HTD-DMCG shows strong target enhancement ability, yielding clear target responses; however, it also produces large activations in certain background regions, which may increase false alarms. HTD-Mamba achieves generally stable detection results, but the separation between target and background responses is relatively weak, with target activations remaining less pronounced. In contrast, our proposed SBSNet shows the best visual detection results across all five datasets. By simultaneously considering spatial and spectral features, the SBSNet effectively suppresses background noise and clearly detects target locations in complex backgrounds, achieving a pronounced background-target separation effect.

G. Ablation Analysis

To systematically analyze the contribution of each key component in the SBSNet, we conduct comprehensive ablation experiments. Specifically, we evaluate: 1) the SPSS; 2) the AESGS; 3) the LSFFM; 4) the CAFTS; 5) the CAMB; and 6) the introduction of the pseudolabel map as auxiliary supervision. Both quantitative and qualitative results are reported to assess their impact on detection performance and training stability. The ablation results are summarized in Tables IV–VI, as well as in Figs. 7–10.

1) *Effectiveness Analysis of the SPSS*: To assess the effectiveness of the SPSS in improving the purity of target sample sets, we perform both qualitative and quantitative analyses. As shown in Fig. 7, the target samples extracted by the SPSS are more consistent with the ground truth than those obtained by HTD-DMCG, indicating higher sample purity and better spectral representativeness. This is because the SPSS avoids selecting a fixed number of extreme-similarity pixels and preserves richer spectral variations in the original HSI. We further evaluate the transferability of the SPSS by integrating the constructed sample sets into the HTD-DMCG framework. The results in Table IV show consistent improvements in $AUC_{(P_d, P_f)}$ across all datasets.

TABLE IV
EVALUATION OF THE TRANSFERABILITY OF THE SPSS ON THE HTD-DMCG
FRAMEWORK ACROSS FIVE DATASETS

Datasets	Metrics	HTD-DMCG	HTD-DMCG + SPSS
SanDiego-1	$AUC_{(P_d, P_f)} \uparrow$	0.9970	0.9991 ($\uparrow$ 0.211%)
	$AUC_{(\tau, P_d)} \uparrow$	0.9135	0.9996 ($\uparrow$ 9.425%)
	$AUC_{(\tau, P_f)} \downarrow$	0.0108	0.0103 ($\downarrow$ 4.630%)
Airport-2	$AUC_{(P_d, P_f)} \uparrow$	0.8207	0.9307 ($\uparrow$ 13.403%)
	$AUC_{(\tau, P_d)} \uparrow$	0.5240	0.7462 ($\uparrow$ 42.405%)
	$AUC_{(\tau, P_f)} \downarrow$	0.0237	0.0414 ($\uparrow$ 9.965%)
Beach-2	$AUC_{(P_d, P_f)} \uparrow$	0.7588	0.9819 ($\uparrow$ 29.402%)
	$AUC_{(\tau, P_d)} \uparrow$	0.6814	0.9130 ($\uparrow$ 33.990%)
	$AUC_{(\tau, P_f)} \downarrow$	0.0879	0.0819 ($\downarrow$ 7.4684%)
Urban-2	$AUC_{(P_d, P_f)} \uparrow$	0.9990	0.9991 ($\uparrow$ 0.010%)
	$AUC_{(\tau, P_d)} \uparrow$	0.9955	0.8963 ($\downarrow$ 9.965%)
	$AUC_{(\tau, P_f)} \downarrow$	0.0124	0.0019 ($\downarrow$ 84.677%)
Qingpu-1	$AUC_{(P_d, P_f)} \uparrow$	0.9999	1.0000 ($\uparrow$ 0.010%)
	$AUC_{(\tau, P_d)} \uparrow$	1.0000	0.8751 ($\downarrow$ 12.490%)
	$AUC_{(\tau, P_f)} \downarrow$	0.0015	0.0003 ($\downarrow$ 80.000%)

The bold values represent the best results.

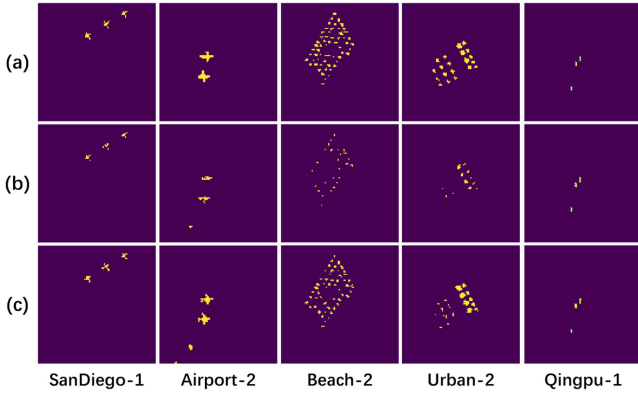


Fig. 7. Visualization of target pixel sets on five benchmark datasets. (a) Ground truth. (b) HTD-DMCG. (c) SBSNet (ours). SBSNet produces target pixel sets that are more consistent with the ground truth.

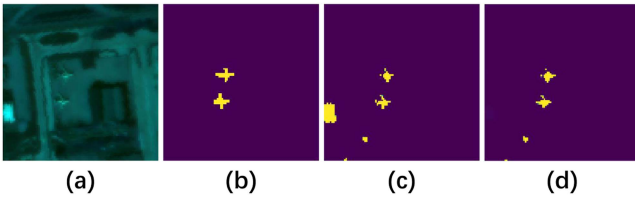


Fig. 8. Visual comparison of detection results before and after applying the proposed CAFTS. (a) Pseudocolor image. (b) Ground truth. (c) Detection result without the CAFTS. (d) Detection result with the CAFTS.

2) *Effectiveness Analysis of the AESGS*: To explore the effectiveness of the AESGS, we conduct experimental analyses on the SBSNet with and without the AESGS. The role of the AESGS is to generate diverse background and target samples, thereby addressing issues related to the scarcity of labeled samples and class imbalance. As shown in Table V, the overall detection accuracy $AUC_{(P_d, P_f)}$ and target detection accuracy $AUC_{(\tau, P_d)}$ improve after the integration of the AESGS, indicating its effectiveness in enhancing the overall performance

of the network. The observed decline in $AUC_{(\tau, P_f)}$ can be attributed to the fact that prior to using the AESGS, the number of background samples is significantly higher than that of target samples. Consequently, the network is more inclined to learn features associated with the background, resulting in better performance in background suppression. From the box plots in Fig. 9, it can be observed that after the integration of the AESGS, the median of the target detection values is closer to 1, while also ensuring effective background suppression.

3) *Effectiveness Analysis of the LSFFM*: To investigate the importance of the LSFFM in enhancing network performance, we conduct experimental analyses on the SBSNet with and without the LSFFM. The purpose of LSFFM is to improve the feature representation of pixels in the HSI by integrating local spectral and spatial information, thereby increasing the accuracy and reliability of target detection. As demonstrated in Table V and Fig. 9, after the integration of the LSFFM, the target detection performance of the SBSNet has improved, and the ROC curve shows better results compared to before, confirming the effectiveness of the LSFFM.

4) *Effectiveness Analysis of the CAFTS*: Based on Section II-D, we propose the CAFTS for the MSSAE. To explore the effectiveness of this strategy, we conduct ablation experiments. background and target samples are clustered through ranking to provide high-quality training data for the AEW-Loss, which helps the MSSAE learn key features more efficiently during training. As shown in the experimental results in Table V, the introduction of this strategy enhances overall detection accuracy $AUC_{(P_d, P_f)}$ and target detection accuracy $AUC_{(\tau, P_d)}$. To provide an intuitive understanding of the effect of the CAFTS on easy and hard samples, Fig. 8 presents a qualitative comparison before and after applying the CAFTS. As shown in Fig. 8(c), a challenging background region in the lower-left area of the Airport-2 scene produces strong false responses without the CAFTS. After applying the CAFTS, this region is effectively suppressed, while target responses remain well preserved. This comparison indicates that the CAFTS enhances discrimination on hard samples without compromising performance on easy samples.

5) *Effectiveness Analysis of the CAMB*: To verify the importance of the CAMB in improving network performance, we conduct an experimental analysis comparing the SBSNet with and without the CAMB. The CAMB enhances feature representation by using attention mechanisms to highlight the most critical features. As shown in the experimental results in Table V, the introduction of CAMB significantly improves the $AUC_{(P_d, P_f)}$ metric and achieves a well-balanced performance between target detection and background suppression, as evidenced by the improvement in AUC_{SNPR} value. As can be seen from the AUC curve in Fig. 9, after the integration of CAMB, the overall detection AUC curve is at the top of all the curves, while the background suppression curve is at the bottom, confirming the effectiveness of the CAMB. To better understand the internal design of CAMB, we conduct subcomponent ablation studies by decomposing the CAMB into the SE and TE blocks. The results on five datasets are reported in Table VI. As shown in the table, different combinations of SE and TE

TABLE V
 QUANTITATIVE DETECTION RESULTS OF ABLATION ANALYSIS FOR THE AIRPORT-2 DATASET

Method	AESGS	LSFFM	CAFTS	CAMB	$AUC_{(P_d, P_f)}$	$AUC_{(\tau, P_d)}$	$AUC_{(\tau, P_f)}$	AUC_{OA}	AUC_{SNPR}	Params	FLOPs
SBSNet-1					0.9235	0.6164	0.0093	1.5306	66.5626	0.0582	0.0019
SBSNet-2	✓				0.9556	0.6311	0.0140	1.5727	45.0786	0.0582	0.0019
SBSNet-3	✓	✓			0.9787	<u>0.6630</u>	0.0148	1.6269	44.9471	0.0582	0.0019
SBSNet-4	✓	✓	✓		0.9822	0.6836	0.0198	1.6460	34.5227	0.0582	0.0019
SBSNet-5	✓	✓		✓	<u>0.9904</u>	0.6555	0.0115	1.6344	57.0522	0.0917	0.0034
SBSNet	✓	✓	✓	✓	0.9947	0.6512	0.0032	<u>1.6427</u>	206.7238	0.0917	0.0034

The bold denotes the best, and underlined denotes the second-best.

 TABLE VI
 ABLATION STUDY OF THE PROPOSED CAMB MODULE, EVALUATED BY $AUC_{(P_d, P_f)}$ ↑, PARAMETERS (M) ↓, AND FLOPS (G) ↓

Module		SanDiego-1			Airport-2			Beach-2			Urban-2			Qingpu-1		
SE	TE	AUC	Params	FLOPs	AUC	Params	FLOPs	AUC	Params	FLOPs	AUC	Params	FLOPs	AUC	Params	FLOPs
✓		<u>0.9995</u>	0.0623	0.0019	0.9784	0.0643	0.0019	<u>0.9960</u>	0.0628	0.0019	<u>0.9929</u>	0.0646	0.0019	<u>0.9998</u>	0.0701	0.0019
	✓	0.9985	0.1316	0.0055	0.9925	0.1336	0.0055	0.9949	0.1321	0.0055	0.9889	0.1339	0.0055	0.9989	0.1394	0.0055
✓	✓	0.9997	<u>0.0896</u>	<u>0.0034</u>	0.9947	<u>0.0917</u>	<u>0.0034</u>	0.9971	<u>0.0901</u>	<u>0.0034</u>	0.9991	<u>0.0919</u>	<u>0.0034</u>	0.9999	<u>0.0974</u>	<u>0.0034</u>

The bold denotes the best, and underlined denotes the second-best.

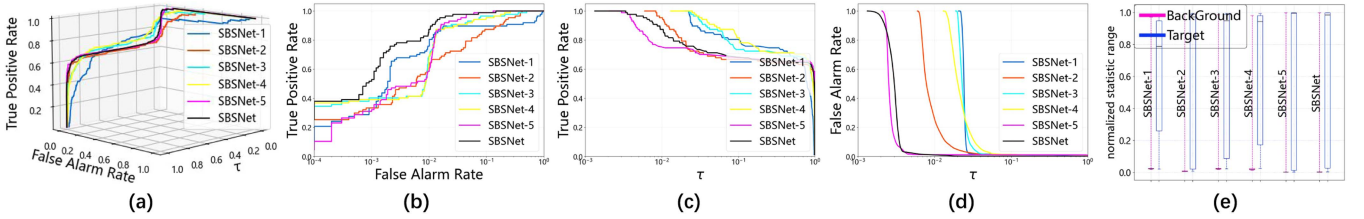


Fig. 9. (a)–(e) ROC curves and box plots of ablation analysis for the Airport-2 Dataset.

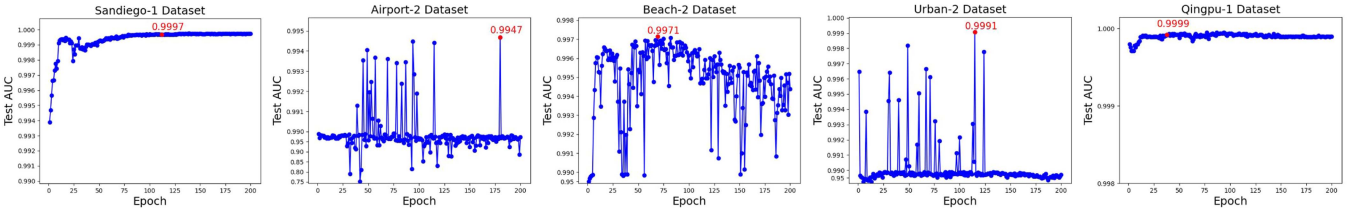


Fig. 10. Variation of testing AUC values across epochs for different datasets, with red dots representing the best results saved under the supervision of the pseudolabel map D_{binary} .

lead to distinct tradeoffs between detection performance and computational cost. Notably, the configuration that integrates both SE and TE consistently achieves the best $AUC_{(P_d, P_f)}$ across all datasets, while maintaining moderate parameters and FLOPs. This demonstrates that integrating SE and TE enables the CAMB to achieve a favorable balance between detection accuracy and computational efficiency.

6) *Effectiveness Analysis of Introducing the Pseudolabel Map*: To validate the effectiveness of introducing the pseudolabel map as supervisory information in improving network performance, we visualize the variation of testing AUC values across epochs during the training process, as shown in Fig. 10. The results shown in the figure demonstrate that, without the pseudolabel map, the test results fluctuate significantly over multiple training rounds, leading to a considerable difference

between the test results of the final training epoch and the optimal results. However, by introducing the pseudolabel map as supervisory information, the model can be guided to preserve better training parameters during the training process.

V. CONCLUSION

To enable the HTD model to have superior background and target separation capabilities, we propose a spatial-spectral background-target separation network for HTD, termed SBSNet. First, we propose the LSFFM to enhance the feature representation capability of raw HSI data. Building on this, we propose the SPPS to construct background and target pixel sets from the unlabeled HSI. In addition, we introduce the pseudolabel map obtained from coarse detection as supervisory information to guide the model in preserving the optimal training parameters

during the training process. Second, we design the MSSAE specifically for HTD, which is utilized for sample generation during data preparation and serves as the backbone network of the SBSNet for feature extraction during training. Third, we propose the CAFTS, aiming to make the MSSAE concentrate on abnormal target and background samples, thereby increasing the separation distance between targets and backgrounds in the feature representation space. To validate the effectiveness of the SBSNet, we conduct extensive experiments on five public HSI datasets. The results show that the SBSNet achieves outstanding performance across various evaluation metrics on all five datasets, demonstrating its superior capability in both target detection and background suppression. In future work, we will investigate cross-domain and cross-sensor evaluations to assess the generalization ability of the proposed SBSNet in more diverse scenarios.

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